Attitude control gyroscope Specification

Document number	SHV-PR-SPEC-0104
Revision	C
Status	APPROVED
Classification	CONFIDENTIAL
Author	Emrys Stagg
Approved by	Cael Merrow
Date	2024-02-22

1. Purpose

This specification defines the performance, interface and verification requirements for the attitude control gyroscope (PR-ACG) of the Propulsion & Station-Keeping on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-PR-05 (WBS 1.5.5), owned by Edda Ingram.

The prime supplier for flight hardware is Chiewab Medical. Responsible engineer: Emrys Stagg.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-PR-DWG-0204	Attitude control gyroscope general arrangement drawing
SHV-PR-BOM-0304	Attitude control gyroscope bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping

4. Description

Propulsion keeps the station in its assigned orbit and attitude. Station-keeping thrusters and sublight ion engines are commanded by the navigation computer, with attitude control gyroscopes for fine pointing.

The attitude control gyroscope is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
PR-ACG-R01	The attitude control gyroscope shall provide a angular momentum of 5.1 GN·m·s.	5.1 GN·m·s	±0.5%	Test
PR-ACG-R02	The attitude control gyroscope shall provide a gyroscope count of 8 ea.	8 ea	±5%	Analysis

5.1 General requirements

ID	Requirement	Verification
PR-ACG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
PR-ACG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
PR-ACG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
PR-ACG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
PR-ACG-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the PR test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-06-21	Initial release.	Emrys Stagg
B	2023-09-17	Clarified verification methods.	Emrys Stagg
C	2024-02-22	Updated requirement values after design review.	Emrys Stagg